

KITEWRIGHT RANGE · DESIGN DECISION RECORD

Kitewright Range TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	KWR-DDR-001	Version	0.1
Status	Draft	Date	2026-10-03
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kitewright-range		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	TRL 2 review decisions D1 to D10, decided under Amish's 2026-10-03 pre-approval

0001: TRL 2 review decisions

- **Date:** 2026-10-03
- **Status:** accepted under Amish's pre-approval. Amish, 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." and, for this batch, "Proceed with the remaining 15 scaffolds".

■ Context

The scaffold (KWR-PRB-001 v0.1) left five open questions: the reference firmware, the wing span and aspect ratio, one or two propeller sets, whether 1 kg of payload is enough for LakeWatch with its winch, and the first co-design partner. Populating the concept to TRL 2 needed further choices on layout, battery, materials and safety. Each was decided as recommended under the pre-approval above. Requirements that the design does not meet are not decided here; they are open decisions for Amish in `docs/06-design-decisions.md`.

■ Options considered and decisions

Table 1. Decisions

#	Question	Options considered	Decision	Reason
D1	Reference firmware	PX4; ArduPilot	PX4 quadplane (VTOL standard) with published parameters; an ArduPilot QuadPlane set may follow	Kitewright Core targets upstream PX4 with no source changes
D2	Wing span and aspect ratio	2.2 m one piece; 2.5 m in two panels; 3.0 m in three panels	2.50 m in two 1.18 m panels on a carbon joiner, NACA 2412, root chord 310 mm, tip 250 mm, aspect ratio 8.9	Largest span whose panels fit a 1.3 m case (R8) with one joint
D3	One or two propeller sets	One set; a low-altitude and a high-altitude set	One set of 20 x 6.5 in propellers for all altitude bands	At sea level they hover at 46 % of maximum thrust, which is efficient enough; one set avoids a wrong-set crash
D4	LakeWatch sampling winch	On Range; on Lift only	1 kg payload on the Core mount for Range; the sampling winch belongs on Kitewright Lift	A winch needs a hover over one point, which is Lift's job
D5	Layout of the cruise motor and tail	Pusher at the tail with a twin-boom tail; tractor at the nose with a tail boom	Tractor cruise motor with a folding propeller at the nose; conventional tail on a carbon tail boom; lift booms on pylons below the wing	Keeps the lift booms short enough for the cases and the rear rotor wash off the tail
D6	Battery	One or two ColdCell packs; Li-ion or LiPo	Two ColdCell 6S3P Li-ion 21700 packs in parallel, one either side of the wing joiner	Li-ion gives the most energy per kilogram for cruise; two packs share hover current
D7	Materials	Moulded composite; foam, glass, lite-ply and printed parts	Hot-wire cut foam with glass skins, lite-ply fuselage, carbon tubes, printed ASA fittings	Buildable with hobby tools (R10) and repairable in the field
D8	First co-design partner and region	State disaster authority; university glaciology group; mountain NGO	First candidate to approach: the Sikkim State Disaster Management Authority, for lake sites in Sikkim, India; second candidate: a university glaciology group already flying Himalayan lake surveys. Not agreed with either	Sikkim has the high-risk lakes and the 2023 South Lhonak outburst; the authority holds the sites and permissions
D9	Safety approach for powered tests (conservative)	Free hover from the first power-up; staged tests	Staged: motors run with propellers off first; first hovers tied down to ground anchors inside a 15 m keep-out zone; a removable arming plug and a lockable arming switch; no propeller fitted on the bench; packs charged only between 0 and 45 °C pack temperature in a fire-resistant box. Relax the tie-down only after 10 logged tied hovers with no fault, and the keep-out only with the partner's site rules	Large propellers and lithium packs are the main hazards (see safety section)

#	Question	Options considered	Decision	Reason
D10	Failsafes (conservative)	Return and land; land in place	Loss of link or low battery: return to launch on the wing, then land vertically; land in place if the battery reaches the reserve; geofence around the survey area	Remote terrain makes a controlled return safer than an uncontrolled landing

■ Consequences

- The constructable design (KWR-DDR-002) and the calculations (KWR-CAL-001) follow D1 to D10.
- D3 is to be confirmed by AltiRig thrust data at 5,000 m density; D6 depends on the ColdCell format, listed as a cross-repo assumption in `docs/REVIEW.md`.
- D8 records candidates only; no partner has been contacted.